Readme for the UrbanTALES datasheet

Below is a table explaining the meaning of columns in the UrbanTALES.csv file included in https://urbantales.climate-resilientcities.com/, which details metadata of all cases as well as canopy-averaged flow and turbulent properties for each case. Detailed description of the datasets can be found in a preprint paper: https://eartharxiv.org/repository/view/8379/.

Variables	Description [unit]	Reference
NameI	Internal case name	\
NameE	External case name	\
Config	Configuration	\
WD	Wind direction relative to the x-axis, which corresponds to the first dimension in the x–y coordinate system.	\
dxdy	Horizontal resolution [m]	\
City	City name for the case	\
State	State name for the case	\
Country	Country name for the case	\
Longitude	Longitude of the upper left case boundary	\
Latitude	Latitude of the upper left case boundary	\
u_{τ}	Prescribed friction velocity [m/s]	\
x-y	Domain size [m]	\
x/y	Domain ratio [-]	\
lp	Plan area density [-]	
lf	Frontal area density [-]	
gammaH	Alignedness normalized by building height [-]	Lu et al, 2023 JAMES
gamma	Alignedness [-]	Lu et al, 2023 JAMES
hmean	Mean building height [m]	
hstd	Standard deviation of building height [m]	

hmax	Maximum building height [m]	
hmin	Minimum building height [m]	
H/h	Domain height over mean building height [-]	
SAR	Case configuration (Staggered, Aligned, Realistic)	
Below are canopy-averaged (from ground to the mean building height), where the same variable name can be found in the vertical profile files for each case. Or single values such as displacement height and aerodynamic roughness length that reflect the characteristics of the whole urban canopy.		
Duw	Dispersive momentum flux [m^2 s^2]	Lu et al, 2024 GMD
Dvw	Dispersive momentum flux [m^2 s^2]	Lu et al, 2024 GMD
Tuw	Turbulent momentum flux [m^2 s^2]	Lu et al, 2024 GMD
Mom	Totoal momentum flux (Tuw+Duw) [m^2 s^2]	Lu et al, 2024 GMD
Tvw	Turbulent momentum flux [m^1 s^2]	
Twc	Turbulent scalar flux [m/s]	Lu et al, 2024 GMD
TKE	Turbulent kinetic energy [m^2/s^2]	Lu et al, 2024 GMD
U	Wind speed [m/s]	`U = <np.sqrt(u**2+v**2)>`
u	Streamwise velocity [m/s]	
v	Spanwise velocity [m/s]	
vel	Velocity [m/s]	`vel = np.sqrt(<u>**2+<v>**2)`
stdS	Standard deviation of passive scalar [-]	Lu et al, 2024 GMD
d	Displacement height [m]	Kanda 2013 BLM
z0	Roughness length [m]	
Below are related to the turbulent kinetic energy budget, Giometto et al, 2016 BLM		
sgsR	Subgrid-scale dissipation using the residue	

	method [m^2/s^2]	
sgsM	Subgrid-scale dissipation using the modelled method [m^2/s^2]	Reference
turbT	Turbulent transport of TKE [m^2/s^2]	
wakeT	Wake transport of TKE [m^2/s^2]	
presT	Pressure transport of TKE [m^2/s^2]	
shearP	Shear production [m^2/s^2]	
wakeP	Wake production [m^2/s^2]	
formP	Form-induced production [m^2/s^2]	
cdeq	Drag coefficient [-]	Nazarian et al., 2020 GMD
Below are pedestrian-level (sampled at 1.75m) values		
pedU	Pedestrian-level wind speed [m/s]	
pedTKE	Pedestrian-level TKE [m^2/s^2]	
pedvel	Pedestrian-level velocity [m/s]	
pedvelU	Pedestrian-level velocity over pedestrian-level wind speed	
pedUstd	Pedestrian-level standard deviation of wind speed [m/s]	
pedUmax	Pedestrian-level maximum value of wind speed [m/s]	
pedS	Pedestrian-level passive scalar [-]	
pedSstd	Pedestrian-level standard deviation of passive scalar [-]	
pedSmax	Pedestrian-level maximum value of passive scalar [-]	
Km	Turbulent diffusion coefficient [m^2/s]	Nazarian et al., 2020 GMD
ugrad	Vertical gradient of streamwise velocity [s^-1]	Nazarian et al., 2020 GMD

cklk	Turbulent length scale [m]	Nazarian et al., 2020 GMD
lece	Turbulent length scale [m]	Nazarian et al., 2020 GMD
cmu	Turbulent viscosity [-]	Nazarian et al., 2020 GMD
rfU	Roof top wind speed [m/s]	
WVR	Wind velocity ratio [-] (Wind speed at the maximum building height vs pedestrian-level wind speed)	Ng 2009, BAE